

Supplementary advice for writing introductions

This document will provide you with a few extra principles to consider when writing introductions. If any of them don't make sense to you, please let me know so I can provide additional explanation or examples.

1st principle

Make the start of the opening as wide as it needs to be and no wider (i.e. begin with topics that are as basic as necessary but not more basic)

This principle applies only if you are writing your mini-paper about a real, complex topic. If you are writing a fictional mini-paper about a simple topic, you can largely ignore it; it probably won't be possible for you to apply this principle,

If you don't understand what "wide" means in this context, review the idea of introductions having an "hourglass" shape from lesson 2.

As wide as needed and no wider

As discussed in lesson 2, the start of the opening should address relatively broad, simple topics to which the reader can easily relate, so that they are able to understand the context of the paper and believe the paper to be important.

Note the word “relatively”: **relatively** broad and simple. You are not trying to write things that are really basic, as though you are writing for a small child; you are trying to pick topics that your educated adult reader finds familiar.

How broad and simple the open topics should be really depends on the reader. If you were writing in *Nature* or *Science* (!) they might be a bit broader, as these journals have readers from all disciplines. If you were writing in a specialist journal for your sub-discipline, they might be a bit narrower, as you might expect all of the readers to already be more expert in the area under discussion.

Your mini-paper is going to be read by me, so consider: I used to be a chemist, a long time ago, and now I’m an English teacher. So the topics at the start of your introduction need to be suitable for someone who has a science background but isn’t a specialist in your particular field. This probably means being more basic than if you were writing for a journal in your discipline – but not as basic as you would be if you were writing for someone with no scientific education.

A common mistake: making the start of the opening too basic/wide

It's easy to start the opening at too low a level, trying to give too much context, telling the reader too many basic things that they already know.

It's fine to do this in the first draft, because it helps you to organise your own ideas, but you should edit it out of the final version.

Example of this mistake from a student mini-paper:

“Humanity’s interest in lives has been universal and enduring. Humans are driven to explore the unknown, discover new worlds, push the boundaries of our limits, and then push forward. The intangible desire to cure aging and diseases, add decades to our life span, and to lead healthier, happier lives have been has provided benefits to the development of life sciences. As one of the most important technologies to unlock the mysteries of life, molecular cloning plays an essential role in many contemporary areas of modern biology and medicine. Based on isolation of a gene of interest to obtain multiple identical copies of it in vitro, it is a set of experimental methods that are used to assemble recombinant DNAs and to direct their replications within host organisms.”

How many sentences of this could we delete?

Example of this mistake from a student mini-paper:

~~"Humanity's interest in lives has been universal and enduring. Humans are driven to explore the unknown, discover new worlds, push the boundaries of our limits, and then push forward. The intangible desire to cure aging and diseases, add decades to our life span, and to lead healthier, happier lives have been has provided benefits to the development of life sciences. As one of the most important technologies to unlock the mysteries of life, molecular cloning plays an essential role in many contemporary areas of modern biology and medicine. Based on isolation of a gene of interest to obtain multiple identical copies of it in vitro, it is a set of experimental methods that are used to assemble recombinant DNAs and to direct their replications within host organisms."~~

At least 3 sentences are pointless and can be deleted.

If the paper really does relate to one of the ideas addressed in the deleted sentences (e.g. preventing aging) then this can be mentioned in a briefer, more focused way.

Some basic information is important and some is pointless

The following sentence is the first sentence of the introduction of a paper in *Science* about the genetic history of donkeys:

“Domestic donkeys (Equus asinus) have facilitated the movement of goods and people for millennia, enabling trade and transport across a broad spectrum of landscapes (1)”.

Todd, E. T. *et al.* The genomic history and global expansion of domestic donkeys. *Science* **377**, 1172–1180 (2022).

The information provided here is quite basic (wide), but it’s essential for justifying why the work is important:

- Historical importance of donkeys
- Continued usefulness of donkeys now

If donkeys weren’t important, research into their genetic history wouldn’t be important; because donkeys are important, the research matters. So, it’s worth saying these basic things to show that donkeys are important.

Equally, though, the sentence does not give *pointlessly* basic information. It does not begin by defining a donkey, or by explaining the importance of trade and transport.

It begins at the most basic level that is directly relevant to the paper: a very brief explanation of why donkeys are important and interesting.

Some basic information is important and some is pointless

If you're beginning your paper with something basic just to explain something that your reader probably already understands, this is a mistake.

However, if you're beginning your paper with something basic to emphasise that your work links to this basic thing, this is more likely to be a good idea. Work that links closely to basic things is often important.

(Obviously, an 8-link chain of reasoning desperately trying to link your work to something fundamental is not convincing.)



2nd principle

Don't waste the first words of the introduction

A common mistake: wasting the first words

If the first sentence of the donkey paper made this mistake, it might start like this:

“This paper addresses some fundamental questions about domestic donkeys, which have facilitated the movement of goods and people for millennia, enabling trade and transport across a broad spectrum of landscapes.”

The opening words here tell us nothing. “This paper address some fundamental questions about” could be the start of 1,000,000 different papers.

But actually, the donkey paper has a strong beginning. It doesn’t start with pointless words like “We have studied...” or “This paper...”:

~~*“This paper addresses some fundamental questions about domestic donkeys, which have facilitated the movement of goods and people for millennia, enabling trade and transport across a broad spectrum of landscapes.”*~~

If the start of your paper uses such vague language that it could be the start of almost any paper, this is a waste. Use the opening words to give real information (if this is reasonably possible – sometimes, of course, vague words are necessary).

The donkey paper also makes the **subject** of the paper clear immediately.

The first sentence could have been written like this:

“The movement of goods and people has been facilitated by donkeys for millennia, enabling trade and transport across a broad spectrum of landscapes.”

When the first sentence is written like this, what does it suggest the paper is about? The movement of goods and people, or donkeys?

If the reader has to expend effort working out the topic of the paper during the first few sentences, this is not helpful. It means they have less concentration available for following the real argument.

Instead, the real first sentence puts donkeys right at the start:

“Domestic donkeys (Equus asinus) have facilitated the movement of goods and people for millennia, enabling trade and transport across a broad spectrum of landscapes (1).”

It's clear from the first two words that this paper is about donkeys.

It may not always be possible to make the general topic of a paper this clear immediately. If you're writing about a complex topic that needs careful introduction, it may take several sentences to get to your real topic.

The point is that you should try to make your topic clear **as soon as you reasonably can**.

3rd principle

Make the start of the opening concrete

The start of the opening should be concrete for the reader

It's important that the opening starts with something that is concrete rather than abstract.

We build new knowledge on top of old knowledge. It's much easier to understand new ideas if we can see how they are linked to things that we already know.

In other words, give specific details rather than making vague statements.

Example of a concrete opening

Todd et al. does not start *“Domestic donkeys (Equus asinus) have been important to humans for millennia.”*

This would be vague and leave us with the question “How have they been important?”

Instead, the start looks like this and gives us concrete information:

“Domestic donkeys (Equus asinus) have facilitated the movement of goods and people for millennia, enabling trade and transport across a broad spectrum of landscapes (1).”

Example of a non-concrete opening

Here's an example of a real non-concrete opening.

“The increasingly serious problem of resource shortage and environmental pollution has brought about potential threats in the lives of human beings. Therefore, solving this problem has become the focus of public concern.”

Xu, J., Liu, X., Hu, L., Li, Z. & Ma, Y. A novel type-II NiCo-LDH/CeO₂ heterojunction for highly efficient photocatalytic H₂ production. New J. Chem. (2022) doi:10.1039/D2NJ02848D.

We are left wondering:

Which resources?

What sort of pollution?

What threats?

Whose lives?

With regard to public concern: when, where, how much?



Nothing is concrete, nothing is specific, so the start of the opening lacks impact.

4th principle

Make the *challenge* specific

Mistake: vague challenge

Example from a student paper:

“... Machine learning is a solution for this problem. This paper will discuss how machine learning is used in aircraft design.”

It is not really clear what this paper is going to do.

This would be like the challenge of the donkey paper saying:

“Overall, the history of donkeys is unclear. This paper will investigate the origin of donkeys.”

Mistake: vague challenge

A more specific way to express the challenge of the donkey paper would be:

“Overall, this lack of consensus between genetic and archaeological data means that the origin and global spread of donkeys are both unclear.”

And this is what the challenge of the donkey paper actually says:

“Overall, this lack of consensus between genetic and archaeological data means that the geographic and temporal origin of donkeys and whether they were domesticated more than once remain uncertain. The global spread of donkeys is also unclear, as their worldwide patterns of genomic diversity lack extensive characterization.”

You might find this hard to read. In simpler language, it is saying: we don't know where and when donkeys were domesticated (i.e. tamed from wild animals) and we don't know whether they were domesticated once or several times. And we don't know how donkeys spread around the world either. This could be expressed as three questions:

- Where and when did donkeys originate?
- Were they domesticated once or multiple times?
- How did they spread around the world?

Mistake: vague challenge

The questions on the previous slide are quite precise.

This makes them a good challenge: the challenge should tell the reader *precisely* what the paper is going to find out or solve; it should not just indicate a broad area of investigation, like the example from the student paper two slides ago.

If you want an example of some precise challenges in a fictional mini-paper, look at the example fictional mini-paper I sent you (the one about the invention of the roof).

5th principle

Phrasing the challenge

Phrasing the challenge

There are two common forms for a challenge.

It might be phrased as **question**, or it might be phrased as a **need**.

If the **challenge** is a **question**, it might be written as a literal question with a question mark (?) at the end. In some subjects/journals, it is common to finish the introduction by providing a research question or list of research questions.

Equally, though, a question-type challenge does not have to be written as a literal question. Very often, the question is implied rather than asked.

E.g. Williams and Bargh, the paper about warmth we read in week 2, has the following challenge: *“For these theoretical and empirical reasons, we hypothesised that mere tactile experiences of physical warmth should activate concepts or feelings of interpersonal warmth.”*

This is a statement of a hypothesis, not a question – but the question it suggests is clear:

Will tactile experiences of physical warmth (i.e. touching a warm object) activate concepts or feelings of interpersonal warmth?

If the challenge is a **need**, it tends to be presented as an opposition between what we (the research community) already have and the new and better thing that the paper provides. For example:

“To date, efforts to consolidate these data or build new large-scale data sets have covered greenhouse gas (GHG) emissions only (8, 12, 13), agriculture only (13–16), small numbers of products (8, 14–16), and predominantly Western European producers (12–16) and have not corrected for important methodological differences between LCAs (12–16). Here, we present a globally reconciled and methodologically harmonized database on the variation in food’s multiple impacts.”

Reducing food’s environmental impacts through producers and consumers, J. Poore and T. Nemecek, Science, 2018, 360, 6392, 987-992

(This is the challenge of the paper in the *Examining the width of an introduction.pdf* document that’s also on the mini-paper resources page on Blackboard.)

This example splits clearly into *what we already have* and *the new and better thing that this paper provides*:

“[The not-very-good thing that world has already:] *To date, efforts to consolidate these data or build new large-scale data sets have covered greenhouse gas (GHG) emissions only (8, 12, 13), agriculture only (13–16), small numbers of products (8, 14–16), and predominantly Western European producers (12–16) and have not corrected for important methodological differences between LCAs (12–16).* **[The new and better thing that this paper provides:]** *Here, we present a globally reconciled and methodologically harmonized database on the variation in food’s multiple impacts.”*

Reducing food’s environmental impacts through producers and consumers, J. Poore and T. Nemecek, *Science*, 2018, 360, 6392, 987-992

See separate document “Examining the width of an introduction” for more context.

Preview

The challenge may be the end of the introduction.

Equally, in many disciplines, it is common for the introduction to continue after the challenge.

The part of the introduction after the challenge is usually a **preview**: it gives a brief overview of the rest of the paper and its ultimate conclusions. If you phrase the challenge as a **need**, the second part of the challenge, (the new and better thing that the paper provides, see previous slide) is likely to flow very naturally into a preview; indeed, the second part of the challenge may **be** the start of the preview (see example on next two slides).

Example challenge + preview

[This example is from an introduction that talks about climate change and architecture. The paragraph below is the final paragraph of the introduction.]

Roofs are crucial for sheltering humans from rainfall, snowfall, and sunlight, and they are the most climate-sensitive, vulnerable, and exposed parts of buildings (21, 22). Therefore, investigating the historical adjustment of roofs is likely to provide insight into the relationship between architecture design and adaptation to climate change (23). To the best of our knowledge, no previous published study has investigated this relationship on a millennium time scale. In this study, we explore the impacts of climate change on roof modification and people's acclimating behavior from a dwelling perspective based on reconstructed chronosequence of roof pitches of ~200 official architectural remains together with high-resolution reconstructed paleoclimatic data (24, 25) in the northern part of central and eastern (NCE) China. The results suggest that the main factor driving the millennial-scale modification of roofs is the change of extreme snowfall events. This study urges greater scientific and societal attention, as we are facing escalated anthropogenic climate change that calls for affordable, reliable, and resilient climate adaptation strategies, particularly in the Global South, given their limited access to adequate climate adaptation resources. Adapted from Li *et al.*, Change of extreme snow events shaped the roof of traditional Chinese architecture in the past millennium, *Science Advances*, vol 7, issue 37 (2021).

We can view this extract as being a question-type challenge plus a preview of the rest of the paper (the question being something like, “what is the relationship between architecture design and adaptation to climate change on a millennium timescale?”).

Example challenge + preview

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We can also view it as being a **need-type challenge**, split into *the current situation* and *the new and better thing that this paper provides*, with the latter turning quite naturally into a preview of the rest of the paper.

Preview

It's up to you whether or not you include a preview in your mini-paper. I suggest you do whatever you are most comfortable doing, considering your choice of topic and the normal practice of your subject.

If you do include a preview, feel free to write in whatever style your subject normally uses. In some subjects, a preview might be just a few sentences immediately following the challenge; in others, it might be an entire separate subsection with a subtitle like "Our contribution".

Also, if you do include a preview, don't forget to make the challenge clear first.

Further reading on tense in introductions

If you'd like a small amount of further reading on the use of tense in introductions, have a look at this book.

English for Writing Research Papers – Adrian Wallwork

[ProQuest Ebook Central – Reader](#)

p 259, pp 271-273

(This book considers introductions and literature reviews separately, so the relevant information is separated between two chapters.)